

H.W.

Quick Sort

54	12	19	3	8	46	31
0	1	2	3	4	5	6

$n=7$, first=0, last=6

i	j	a[i]	a[j]	a[pivot]
0	6	54	31	54

54	12	19	3	8	46	31
i	j	i	j	i	j	i

$a[j] \rightarrow a[pivot]$

31 $\leftrightarrow$ 54

31	12	19	3	8	46
0	1	2	3	4	5

first=0, last=5

i	j	a[i]	a[j]	a[pivot]
0	5	31	12	31

31	12	19	3	8	46
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1

i

1

1

1

① ①

$$a[i] \leftrightarrow a[\text{pivot}]$$

$$8 \leftrightarrow 31$$

8	12	19	3
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31

46

54

i

j

a[i]

a[j]

a[pivot]

0

3

8

3

8

8	12	19	3
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i

①

j

$$a[i] \leftrightarrow a[j]$$

swap

12 & 3

8	3	19	12
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i

j

i

①

①

$$i > j$$

$$a[j] \leftrightarrow a[\text{pivot}]$$

swap 3 & 8

3	8	19	12
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31

46

54